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Оригинальная статья

Разработка упрощенной шкалы оценки тяжести и упрощенных критериев для назначения системного лечения атопического дерматита методом Дельфи

Чжан, Ли¹; Цао, Шэнсинь²; Дин, Янфэн³; Гэн, Сунмэй⁴; Го, Хао¹; Цзи, Чжао⁵; Цзян, Цицзюнь⁶; Ли, Вэй⁷; Лян, Юньшэн⁸; Мань, Сяюнь⁹; Сун, Чжицян¹⁰; Сунь, И¹¹; Сяо, Жун⁸; Яо, Сюй¹²; Юй, Чуньшуй¹³; Гао, Синхуа¹

Редактор(ы): Го, Лишао

[Информация об авторе](#)

¹Department of Dermatology, The First Hospital of China Medical University, Key Laboratory of Immunodermatology, Ministry of Education, and National Health Commission, Shenyang, Liaoning 110001, China

²Department of Dermatology, The People's Hospital of Yizheng, Yangzhou, Jiangsu 225400, China

³Department of Dermatology, Shanghai Skin Disease Hospital, Tongji University, School of Medicine, Shanghai 200443, China

⁴Department of Dermatology, The Second Affiliated Hospital of Xi'an Jiaotong University, Xi'an, Shaanxi 710004, China

⁵Department of Dermatology, The First Affiliated Hospital of Fujian Medical University, Fuzhou, Fujian 350005, China

⁶Department of Dermatology, Donggang Central Hospital, Donggang, Liaoning 118300, China

⁷Department of Dermatology, Huashan Hospital, Fudan University, Shanghai 200040, China

⁸Department of Dermatology, Second Xiangya Hospital, Central South University, Changsha, Hunan 410011, China

⁹Department of Dermatology, Second Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang 310009, China

¹⁰Department of Dermatology, Southwest Hospital, Army Medical University, Chongqing 400038, China

¹¹Department of Dermatology, Jingzhou Hospital Affiliated to Yangtze University, Jingzhou, Hubei 434000, China

¹²Department of Allergy and Rheumatology, Hospital of Skin Disease (Institute of Dermatology), Chinese Academy of Medical Sciences and Peking Union Medical College, Nanjing, Jiangsu 210042, China

Correspondence to: Xinghua Gao, Department of Dermatology, The First Hospital of China Medical University, 155N Nanjing Street, Heping District, Shenyang, Liaoning 110001, China E-Mail: xhgao@cmu.edu.cn

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Аннотация

Краткое изложение на простом языке

Предыстория:

В настоящее время клиническая оценка тяжести атопического дерматита (АД) и назначение системного лечения остаются проблематичными из-за сложных и трудоемких протоколов обследования. Из-за этой сложности врачи часто полагаются на субъективные клинические суждения, особенно в учреждениях первичной медико-санитарной помощи, что нередко приводит к несвоевременному терапевтическому вмешательству и недостаточно эффективному лечению. Целью проекта была разработка упрощенных стандартизированных инструментов для оценки тяжести АД и принятия решений о назначении системного лечения для повышения клинической эффективности и улучшения результатов лечения пациентов.

Методы:

Первоначальный набор показателей для упрощенной шкалы оценки атопического дерматита был составлен на основе всестороннего анализа литературы и консультаций с экспертами. Затем был проведен двухэтапный электронный опрос по методу Дельфи с участием 16 ведущих дерматологов из Китая, чтобы оценить и доработать эти показатели, а также установить согласованные критерии для назначения системного лечения. Исследование проводилось с ноября 2023 года по июнь 2024 года.

Результаты:

В ходе двухэтапного процесса консенсуса по методу Дельфи мы разработали: во-первых, упрощенную шкалу оценки атопического дерматита, включающую три основных показателя (зуд, кожные поражения и качество жизни) и пять дополнительных показателей (интенсивность зуда, влияние зуда на сон, площадь поражения кожи в процентах от площади поверхности тела (ППТ), комплексная оценка интенсивности кожных поражений и комплексная оценка влияния на повседневную активность). Во-вторых, упрощенные критерии для назначения системного лечения, в том числе: (А) поражение кожи, занимающее более 10 % площади поверхности тела, или поражение менее 10 % площади поверхности тела, затрагивающее чувствительные или хорошо заметные участки, или недостаточный ответ на предшествующее местное лечение; (В) серьезные нарушения сна из-за зуда, требующие снотворных препаратов или приводящие к пробуждению; (С) значительное ухудшение качества повседневной деятельности, в том числе работы, учебы или социальной активности. Системная терапия рекомендовалась при наличии признаков В или С. Упрощенная шкала оценки болезни Альцгеймера и критерии начала системной терапии обладают тремя явными клиническими преимуществами для применения в Китае: (1) высокая практическая ценность благодаря качественным дескрипторам, заменяющим громоздкие количественные показатели; (2)

комплексная оценка с использованием интегрированных параметров, о которых сообщают пациенты и которые оценивает врач; (3) адаптированность к региональным особенностям — это первый инструмент оценки, разработанный на основе консенсуса в Китае. Несмотря на то, что эти новые инструменты уже доказали свою клиническую ценность, для подтверждения их надежности, валидности и клинической применимости необходимы дальнейшие исследования.

Вывод:

Упрощенная шкала оценки атопического дерматита и связанные с ней критерии для назначения системной терапии могут повысить клиническую эффективность и упростить процесс принятия решений китайскими дерматологами при лечении атопического дерматита.

Введение

Атопический дерматит (АД) — распространенное хроническое воспалительное заболевание кожи II типа,^[1] от которого страдают около 7,3 % взрослого населения США (по данным 2019 года) и 7,8 % населения Китая (по данным 2014 года).^[2] Хотя легкую форму атопического дерматита обычно можно эффективно контролировать с помощью смягчающих средств и местной терапии, атопический дерматит средней и тяжелой степени или распространенная форма заболевания представляют собой серьезную проблему, требующую длительного системного лечения и существенно влияющую на качество жизни пациентов из-за частых рецидивов. Таким образом, лечение болезни Альцгеймера по-прежнему остается сложной задачей как для пациентов, так и для медицинских работников.

The 2020 Chinese Guideline for Diagnosis and Treatment of Atopic Dermatitis recommends using the SCORing Atopic Dermatitis (SCORAD) index to classify AD severity into mild, moderate, and severe categories, supporting clinical decision-making and guiding the initiation of systemic treatments.^[3] By 2022, expert recommendations on treat-to-target in the systemic treatment of moderate-to-severe AD advocated treatment regimens based on measurable improvements in validated indices, including the Eczema Area and Severity Index (EASI), SCORAD, peak pruritus numerical rating scale (PP-NRS), Dermatology Life Quality Index (DLQI), and patient-oriented eczema measure (POEM) scores during both induction and maintenance phases.^[4,5] Such a targeted approach highlights the essential role of severity assessment in diagnosing AD, developing treatment plans, and evaluating therapeutic outcomes.

Several validated AD assessment scales are currently available and are frequently used as endpoints in clinical trials. However, existing scales exhibit several notable limitations: (1) They often individually measure specific domains such as symptom severity, pruritus, quality of life, or long-term disease control, yet no single instrument comprehensively captures all relevant treatment outcomes. (2) Widely used scales, including SCORAD^[6] (introduced in 1997 by the European Task Force on Atopic Dermatitis) and EASI^[7] (developed in 1998), involve complex calculations that limit their practicality in routine clinical practice, especially within primary care settings. This complexity can result in subjective assessments and potentially suboptimal management of AD. (3) Furthermore, both EASI and SCORAD present significant limitations when applied to Chinese patients with AD. A notable challenge is the accurate assessment of erythema (skin redness), a crucial element in these indices, especially in darker skin tones common among Chinese populations, potentially leading to underestimated severity.^[8] In addition, these scales may inadequately reflect the profound impact of lesions located on culturally and socially important areas such as the face, hands, and feet, suggesting the need for assessment tools that fully address these essential aspects of disease burden.

To enhance clinical application, improve disease management efficiency, and facilitate decision-making, particularly regarding initiation of systemic treatment—Chinese dermatologists initiated the Practice-based AD Advance Leadership (PADDLE) project. This initiative aims to develop a Simplified AD Assessment Scale that integrates critical evaluation dimensions while prioritizing simplicity and practicality. The primary goal is to create a scale with enhanced reliability and usability compared with existing tools, making it accessible and effective for dermatologists at all healthcare levels in China, ranging from primary care providers to advanced specialists.

Moderate-to-severe AD frequently requires systemic therapies, such as immunosuppressants, glucocorticoids, biologics, and small-molecule targeted therapies, since topical treatments alone often prove inadequate. The expert recommendations on treat-to-target in the systemic treatment of moderate to severe AD suggest initiating systemic therapy when patients fulfill three distinct standards: (1) a relatively objective standard by PGA, EASI, and SCORAD; (2) a subjective standard by DLQI, VAS, or NRS; and (3) an insufficient treatment response.^[4] In 2021, the Atopic Dermatitis Group of the Portuguese Society of Dermatology and Venereology recommended JAKi or biologics for patients with an EASI score greater than 16, or for those with EASI scores below 16 who meet at least one additional criterion: (1) localized on face, hands or genitals; (2) pruritus NRS >7; (3) sleep disturbances with NRS >7; (4) DLQI >10.^[9] Despite their structured nature, these existing criteria are complex, potentially delaying systemic treatment initiation and causing inconsistencies in clinical decision-making across practitioners.

To streamline systemic treatment decision-making in AD, the PADDLE project drew inspiration from the 2019 International Psoriasis Council (IPC)'s simplified classification. Based on the Delphi method,^[10] this framework identifies candidates for systemic therapy by requiring only one of the following criteria to be met: (1) body surface area (BSA) involvement >10%; (2) involvement of special anatomical areas; or (3) failure of topical therapy. Unlike traditional classifications into mild, moderate, and severe categories, this pragmatic approach has improved efficiency in clinical decision-making.

Accordingly, to simplify AD severity assessment and establish practical criteria for initiating systemic treatment, the present study used expert insights through a Delphi exercise.

Methods

The study was conducted from November 2023 to June 2024, following five key steps [Figure 1]. Step 1 is the construction of the candidate indicator pool. Step 2 is the composition of the questionnaire for the Delphi survey. Steps 3 and 4 are the electronic Delphi (eDelphi) round 1 and round 2. Step 5 is the expert review of the eDelphi result.



Figure 1:

Schematic representation of the project's workflow, highlighting the five major steps, construction of indicator pool, composition of the questionnaire for Delphi survey, two rounds of electronic Delphi survey and finally, expert review of survey results. AD: Atopic dermatitis.

Organization of the expert panel

The recommended number of panelists for Delphi surveys, according to Turoff,^[11] is between 10 and 50. Experts were recruited nationwide between December 2023 and January 2024 based on the following criteria: (1) at least ten years of clinical experience in dermatology; (2) holding the professional title of chief physician or above; (3) preferably with prior experience in Delphi studies; (4) broad geographical distribution; (5) inclusion of both primary care dermatologists and tertiary-level specialists to ensure a diversity of perspectives; (6) demonstrable professional engagement in AD management; and (7) willingness and consent to participate in the study.

Construction of the candidate indicator pool for the Simplified AD Assessment Scale and Simplified Criteria for Initiating AD Systemic Treatment

The candidate indicator pool was developed through a four-step process: Step 1: Systematic organization of global AD-related assessment scales and guidelines into a three-tiered indicator framework, where the third-level indicators elaborate on the severity levels of the second-level indicators (mild, moderate, and severe). Step 2: Integration of additional indicators derived from real-world clinical practice. Step 3: Elimination of redundant items and exclusion of patient-subjective indicators. Step 4: Review and refinement by the expert panel, including the addition, deletion, or modification of indicators as necessary.

Questionnaire development

Development of the eDelphi round 1 questionnaire

The eDelphi round 1 questionnaire was developed based on the candidate indicator pool for the Simplified AD Assessment Scale and Simplified Criteria for Initiating AD Systemic Treatment. It consisted of three sections, as detailed in Supplementary File 1, <https://links.lww.com/CM9/C626>.

Development of eDelphi round 2 questionnaire

Based on the results of round 1, the eDelphi round 2 questionnaire [Supplementary File 2, <https://links.lww.com/CM9/C627>] was developed using the following procedures:

- (1) Elimination of first-level and second-level indicators receiving low importance scores in round 1.
- (2) Removal of third-level indicators receiving low agreement scores regarding AD systemic treatment initiation.
- (3) Integration of expert feedback gathered during round 1.
- (4) Inclusion of statistical metrics, including mean scores and the proportion of full scores or scores above 4, from round 1.

The eDelphi round 2 questionnaire retained the same three-section structure as round 1.

Two rounds of eDelphi survey

The first eDelphi round was conducted in March 2024, with questionnaires distributed through a secure online survey platform (Medlive). Medlive's role was limited to technical programming – converting the Delphi questionnaire into a secure online format.

Panelists were given 3 days to complete the survey anonymously and independently. Before round 2 in April 2024, participants received a summary of round 1 scores and comments for reference.

In total, two eDelphi rounds were completed. Panelists were also consulted for minor revisions throughout the process, as necessary.

Analysis and screening of first-level, second-level, and third-level indicators

Quantitative data from the eDelphi surveys were analyzed using SPSS 20.0 (IBM Corp, Armonk, USA). Analytical dimensions included mean scores and coefficients of variation (CVs), among others. The mean scores of the third-level indicators were used to evaluate consensus among experts on initiating systemic treatment. The CV was used to assess consistency among expert opinions, with lower CV values indicating higher consistency.

The criteria for retaining or excluding indicators are generally more lenient in the initial round to accommodate a broader range of items for discussion and feedback. This inclusive strategy ensures that potentially valuable items are not prematurely excluded. Existing Delphi literature commonly uses a combination of mean scores and CVs to determine item retention. In the first round, retention thresholds are often set at a mean score between 3 and 4, and a CV between 0.2 and 0.3.^[12–16] These parameters aim to balance consensus with variability, ensuring items are neither overly controversial nor universally accepted without proper scrutiny.

As the Delphi process advances, the criteria for retention become more stringent. This is because items remaining in later rounds have already undergone preliminary evaluation and are expected to achieve greater expert consensus. Consequently, mean score thresholds are typically increased to between 4 and 4.5, with acceptable CVs close to 0.2.^[12,16] This adjustment ensures that retained items receive strong expert support and exhibit minimal disagreement, enhancing the reliability of the study outcomes. By applying progressively stricter criteria across Delphi rounds, the indicator list can be effectively refined to ensure that the final selection accurately reflects expert consensus while minimizing variability. This structured methodology contributes to the development of robust and credible outcomes in Delphi-based research.

Therefore, in our study, indicator screening exclusion criteria for Simplified AD Assessment Scale (by first-level and second-level indicator importance score) are: for eDelphi round 1, a mean score <3.5 or a CV >0.25; for eDelphi round 2, a mean score <4.4 or a CV >0.13.

Indicator screening exclusion criteria for Simplified Criteria for Initiating AD Systemic Treatment (by third-level indicator agreement score): for round 1, a mean score <3.5 or a CV >0.25; for round 2, a mean score <4 or a CV >0.2 (if a second-level indicator was removed, its corresponding third-level was also removed).

Positive coefficient

The positive coefficient is a critical indicator that reflects experts' engagement and enthusiasm in the eDelphi process. It is defined as the recovery rate of valid questionnaires, calculated by dividing the number of valid responses by the total number of distributed questionnaires.^[17] A questionnaire was deemed invalid if it (1) was completed in under 30 seconds with all answers identical, or (2) had more than 10% of its questions unanswered. A recovery rate exceeding 60% is typically interpreted as a high level of engagement among panelists.^[18]

Expert authority coefficient

The expert authority coefficient (Cr) reflects the authority level of participating experts. It is based on their familiarity with the study content and the basis of their judgments. Cr is calculated as the average of the familiarity coefficient (Cs) and the judgment coefficient (Ca), according to the formula: $Cr = (Ca + Cs)/2$.

The judgment coefficient (Ca) is determined by aggregating influence scores across four domains: clinical practice experience, theoretical analysis, knowledge of domestic and international literature, and intuitive judgment. Each domain is assigned a corresponding influence score [Supplementary Table 1, <https://links.lww.com/CM9/C625>].

The familiarity coefficient (Cs) includes five levels, each assigned a specific score based on self-reported familiarity with the subject matter [Supplementary Table 1, <https://links.lww.com/CM9/C625>].

Coordination degree of expert opinion

The coordination among expert opinions was evaluated using the CV and Kendall's coefficient of concordance (W). The CV ranges from 0 to 1, with lower values indicating stronger coordination among expert opinions. Similarly, the W coefficient ranges from 0 to 1, with higher values representing greater agreement.^[19]

Expert review

Experts were invited to participate in group discussions to review and revise the outcomes of the two eDelphi rounds. When necessary, unresolved items were subjected to additional voting.

Results

Demographic features of the experts

The expert panel consisted of 16 experienced dermatologists from across China. Detailed demographic characteristics are provided in Supplementary Table 2, <https://links.lww.com/CM9/C625>.

Development of the candidate indicator pool for the eDelphi survey

The process for developing the candidate indicator pool is illustrated in Supplementary Figure 1, <https://links.lww.com/CM9/C625>. Initially, a comprehensive literature review was conducted, referencing 16 commonly used AD assessment scales [Supplementary Table 3, <https://links.lww.com/CM9/C625>] and nine relevant guidelines and academic publications [Supplementary Table 4, <https://links.lww.com/CM9/C625>].^[20–25] In addition, several indicators were included based on clinical experience.

Subsequently, second-level indicators were organized following these principles: (1) Merge indicators that assess the same content. (2) Integrate comprehensive indicators with corresponding partial indicators (those assessing only a subset of the content) into unified comprehensive indicators. (3) Merge

subjective indicators (e.g., patient-reported assessments) with objective indicators (e.g., clinician-reported assessments) when they evaluate the same concept.

Finally, two rounds of expert discussions were conducted to vote on the addition or deletion of indicators, with descriptions of the indicators refined. The finalized candidate indicator pool consists of four first-level, 16 second-level, and 48 corresponding third-level indicators, with each second-level indicator encompassing three third-level indicators that describe escalating severity levels [Supplementary Table 5, <https://links.lww.com/CM9/C625>].

Results of eDelphi survey

Positive coefficient of experts

In both rounds of the eDelphi survey, 16 questionnaires were distributed and subsequently recovered. All retrieved questionnaires were valid, resulting in a 100% positive coefficient—indicating a high level of engagement and enthusiasm among participating experts.^[19]

Expert authority

In dermatological research using the Delphi method, an expert authority coefficient (Cr) greater than 0.7 is generally considered satisfactory.^[26] For instance, a 2023 study “Construction of Expert Consensus on Diagnosis and Treatment of Chronic Inducible Urticaria in China” reported a Cr value of 0.92, indicating a high level of expert authority.^[26] In our study, eDelphi round 1 expert authority coefficient (Cr) was 0.91. Familiarity coefficient (Cs) = 0.97 and judgment coefficient (Ca) = 0.85. eDelphi round 2 expert authority coefficient (Cr) was 0.95. Familiarity coefficient (Cs) = 0.93 and judgment coefficient (Ca) = 0.97. Therefore, the experts in our study possessed high authority, and the data obtained were highly reliable.

Coordination degree of expert opinion

The coordination among expert opinions was considered high, based on CV results from both survey rounds [Supplementary Tables 6 and 7, <https://links.lww.com/CM9/C625>]. Kendall’s coefficient of concordance (W) ranged from 0.2 to 0.5 ($P < 0.0001$), indicating statistically significant agreement among panelists [Supplementary Table 8, <https://links.lww.com/CM9/C625>].

Results of indicator screening for the Simplified AD Assessment Scale by first-level and second-level indicator importance score

The eDelphi round 1 survey included four first-level and 16 second-level indicators. Based on predefined exclusion criteria (mean score < 3.50 or $CV > 0.25$), four second-level indicators—“number of itchy body parts”, “trend of itching intensity (alleviated or worsened)”, “itching area”, and “number of body parts that appeared skin lesion”—were excluded from subsequent rounds. Following expert feedback, the indicator “past topical therapy response” was revised into “past topical therapy response (referring to standard topical therapy, defined as the use of topical medication with potency corresponding to the severity of skin lesions)”. The remaining four first-level and 12 second-level indicators were included in the eDelphi round 2 survey [Supplementary Table 6, <https://links.lww.com/CM9/C625>].

In the eDelphi round 2 survey, seven second-level indicators—“impact on emotions”, “relapse history”, “past systemic therapy response”, “atopic dermatitis-related comorbidities”, “number of body parts that appeared skin lesion”, “duration of itch”, and “past topical therapy response (referring to standard topical therapy, defined as the use of topical medication with potency corresponding to the severity of skin lesions)”, along with one first-level indicator, “Others”, met the exclusion criteria (mean score < 4.4 or $CV > 0.13$) and were removed.

The finalized indicators for the Simplified AD Assessment Scale comprised three first-level indicators and five corresponding second-level indicators: “intensity of itch”, “impact of itching on sleep”, “comprehensive evaluation of the intensity of skin lesion signs”, “skin lesion involvement of body surface area”, and “comprehensive evaluation of the impact on daily activities (life/study/work/appearance impact)”.

The final survey results are presented in [Table 1](#), while detailed data from both rounds of the Delphi survey can be found in Supplementary Table 6, <https://links.lww.com/CM9/C625>.

Table 1 - Screening results for Simplified AD Assessment Scale by second-level indicator importance score.✕

Swipe left or right to view entire table.

Indicators entering the Simplified AD Assessment Scale	eDelphi round 1: Importance score of indicators on assessing AD severity		eDelphi round 2: Importance score of indicators on assessing AD severity	
	Mean	CV	Mean	CV
First-level indicator				
A Pruritus	4.80	0.11	5.00	0
B Skin lesion	4.90	0.05	4.90	0.05
C Quality of life	4.60	0.13	4.90	0.07
Second-level indicator				
A2 Intensity of itch	4.70	0.13	5.00	0
A4 Impact of itching on sleep	4.70	0.13	4.90	0.07
B1 Skin lesion involvement of body surface area	4.50	0.11	4.60	0.13
B3 Comprehensive evaluation of the intensity of skin lesion signs	4.60	0.11	4.80	0.09
C1 Comprehensive evaluation of the impact on daily activities (life/study/work/appearance impact)	4.40	0.16	4.40	0.11

AD: Atopic dermatitis; CV: Coefficient of variation.

Results of indicator screening for the Simplified Criteria for Initiating AD Systemic Treatment by third-level indicator agreement score

In the eDelphi round 1 survey, 48 third-level indicators were assessed. Thirteen of these met the predefined exclusion criteria (mean agreement score <3.5 or CV >0.25) and were excluded from subsequent rounds. In addition, following the removal of second-level indicators—"Number of itchy body parts", "trend of itching intensity (alleviated or worsened)", "itching area", "number of body parts that appeared with skin lesion"—the corresponding 12 third-level indicators were also eliminated. As a result, 23 third-level indicators remained for inclusion in the eDelphi round 2 survey [Supplementary Table 7, <https://links.lww.com/CM9/C625>].

In round 2, six additional third-level indicators met exclusion criteria (mean agreement score <4.0 or CV >0.2) and were removed. Furthermore, based on expert feedback, the criterion for systemic treatment initiation was refined: it should be based on patients who have not previously received systemic treatment. Consequently, the indicator "past systemic therapy response" was eliminated. Experts also reached consensus that when both moderate-level (level B) and severe-level (level C) descriptions of a given indicator were highly agreed upon, the severe-level description should be merged into the moderate-level. Accordingly, six such pairs were merged, yielding six refined indicators at the moderate description level [Supplementary Table 7, <https://links.lww.com/CM9/C625>].

Ultimately, the two rounds of the eDelphi survey produced ten items for initiating systemic treatment for AD [[Table 2](#)].

Table 2 - Screening results for Simplified Criteria for Initiating AD Systemic Treatment by third-level indicator agreement score.✕

Swipe left or right to view entire table.

First-level indicator	Second-level indicator	Third-level indicator	eDelphi round 1		eDelphi round 2	
			Mean	CV	Mean	CV
A Pruritus	A1 Duration of itch	B: 6–12 hours of itching every day (1/4–01/2 time of a day)	4.20	0.16	4.10	0.15
	A2 Intensity of itch	B: Need frequent and violent scratch	4.30	0.14	4.30	0.14
	A4 Impact of itching on sleep	B: Affect falling asleep and wake up due to itch	4.10	0.15	4.10	0.12
B Skin lesion	B1 Skin lesion involvement of body surface area	C: >10% body surface area	4.60	0.14	4.60	0.13
	B2 Location of skin lesion (refers to sensitive or visible parts of the body, including perineum, nipples, head and face, neck, hands. Symmetrical sites count as 1, e.g., left and right hand only count as 1)	C: Appeared skin lesion, 2 and more special parts of the body involved (sensitive or visible body sites)	4.30	0.20	4.20	0.18
	B3 Comprehensive evaluation of the intensity of skin lesion signs	C: Very conspicuous lesion	4.70	0.13	4.50	0.16
C Quality of life	C1 Comprehensive evaluation of the impact on daily activities (life/study/work/appearance impact)	C: Always affect daily activities	4.40	0.14	4.50	0.14
	C2 Impact on emotions	B: Large impact	4.10	0.12	4.00	0
D Others	D1 Past topical therapy response	C: Poor response to topical therapy	4.10	0.19	4.50	0.14
	D3 Relapse history	B: 3–6 times of relapse episodes every year	3.70	0.19	4.00	0.16

AD: Atopic dermatitis; CV: Coefficient of variation.

Expert review and results

The finalized Simplified AD Assessment Scale includes five second-level indicators: intensity of itch, impact of itching on sleep, skin lesion involvement of body surface area, comprehensive evaluation of the intensity of skin lesion signs, and comprehensive evaluation of the impact on daily activities.

Severity levels (mild, moderate, and severe) were derived from third-level indicators in the candidate pool [Supplementary Table 5, <https://links.lww.com/CM9/C625>]. An additional category–“none”–was added to correspond to the absence of symptoms.

During expert review, all panelists unanimously agreed on the final indicator selection for the Simplified AD Assessment Scale [Table 3].

Table 3 - Finalized Simplified AD Assessment Scale✕
Swipe left or right to view entire table.

First-level indicator	Second-level indicator	Third-level indicator			
		None	Mild	Moderate	Severe
Pruritus	Intensity of itch	Barely or sometimes itch but no need to scratch	Mild and intermittent scratch	Need frequent and violent scratch	Scratch nonstop violently, lead to broken skin

First-level indicator	Second-level indicator	Third-level indicator			
		None	Mild	Moderate	Severe
	Impact of itching on sleep	Never affect sleep	Occasionally affect falling asleep, never wake up due to itch	Affect falling asleep and wake up due to itch	Cannot habitually fall asleep due to itch and scratch, need sleep aids
Skin lesion	Skin lesion involvement of body surface area	0%	>0%, <5% of body surface area	5–10% of body surface area	>10% of body surface area
	Comprehensive evaluation of the intensity of skin lesion signs (signs include erythema, papule, papulovesicle/exudation/incrustation, lichenification, excoriation, dryness)	No lesion or barely visible lesion	Visible, mild skin lesion	Obvious skin lesion	Very conspicuous skin lesion
Quality of life	Comprehensive evaluation of the impact on daily activities (life/study/work/appearance impact)	Does not or rarely affect daily activities	Occasionally affect daily activities	Frequently affect daily activities	Always affect daily activities

AD: Atopic dermatitis.

In the same review meeting, the 10 remaining third-level indicators for initiating systemic treatment—identified through two rounds of the eDelphi survey—were also presented. Drawing on the International Psoriasis Council’s definition of “candidates for systemic therapy”,^[10] experts reached consensus to further simplify the criteria. They agreed that systemic treatment initiation could be determined by combining several third-level indicators. The expert panel used a three-step process [Supplementary Figure 2, right, <https://links.lww.com/CM9/C625>] to finalize the Simplified Criteria for Initiating AD Systemic Treatment. Consensus was reached that patients should meet “1+1” criteria (A plus one of B or C), with A referring to “Skin lesion involvement of body surface area >10% body surface area, or <10% body surface area but involve special area (sensitive or visible area); failure of past topical therapy”, B referring to “Cannot fall asleep (need sleep aids) or wake up in the night, because of itch”, C referring to “Work/study/socializing activities are severely affected, significantly impaired quality of life”.

Discussion

Strengths and limitations of the Simplified AD Assessment Scale

Numerous AD assessment scales are available, but few offer a comprehensive evaluation across multiple domains. SCORAD, for example, assesses skin lesion involvement, intensity of skin lesion signs, pruritus severity, and sleep disturbance. However, its complexity limits its practicality in routine clinical use, particularly in lower-tier healthcare facilities. To address this limitation, the present study developed a Simplified AD Assessment Scale.

An initial candidate indicator pool—including four first-level, 16 second-level, and 48 third-level indicators—was compiled through literature review and expert consultation. Following two rounds of the eDelphi survey, five second-level indicators were selected to form the finalized scale, encompassing key disease dimensions alongside qualitative severity levels.

Currently, both domestic and international guidelines for assessing AD severity primarily rely on tools such as SCORAD and EASI. For instance, the 2020 Chinese Guideline for the Diagnosis and Treatment of Atopic Dermatitis classifies severity as mild (SCORAD: 0–24), moderate (25–50), and severe (>50).^[3] In contrast, the Guidelines of Care for the Management of Atopic Dermatitis by the American Academy of

Dermatology do not advocate routine use of such scales in clinical practice, noting they were not originally developed for routine severity evaluation.^[3] Instead, it suggests that clinicians ask general questions regarding itch, sleep, daily activity impact, and disease persistence, with available scales used primarily when practical.^[3]

Aligned with this approach, the Simplified AD Assessment Scale focuses on pruritus, skin lesions, and quality of life—domains frequently emphasized in real-world settings. Drawing on expert input, the assessment dimensions were streamlined and refined. The resulting scale integrates both detailed assessments and simplified qualitative severity levels (none, mild, moderate, severe), enhancing clinical feasibility and real-world applicability. A visual comparison of the new scale and existing scales is illustrated in [Table 4](#).

Table 4 - Visual comparison of newly developed scale in our study and existing scales.✂
Swipe left or right to view entire table.

Assessment scales	Evaluation dimensions				Used by
	Skin lesion	Pruritus	Sleep	QoL	
Simplified AD Assessment Scale	√	√	√	√	Physicians
Existing scales					
EASI	√				Physicians
SCORAD	√	√	√		
IGA	√				
BSA	√				
vIGA × BSA	√				
PP-NRS		√			Patients
POEM	√	√	√		
DLQI				√	
ADCT		√	√	√	

AD: Atopic dermatitis; ADCT: Atopic Dermatitis Control Tool; BSA: Body surface area; EASI: Eczema Area and Severity Index; DLQI: Dermatology Life Quality Index; IGA: Investigator Global Assessment; POEM: Patient-oriented eczema measure; PP-NRS: peak pruritus numerical rating scale; QoL: Quality of life; SCORAD: SCORing atopic dermatitis; vIGA: Validated Investigator Global Assessment.

The newly developed scale demonstrates three significant improvements over existing assessment tools. First, in contrast to conventional AD severity scales that predominantly employ quantitative metrics (e.g., EASI, SCORAD), our instrument utilizes qualitative descriptors for key clinical indicators. This approach provides more clinically intuitive evaluation criteria while maintaining diagnostic precision, thereby facilitating rapid severity assessment in routine practice settings. Second, the scale innovatively integrates patient-reported outcomes (particularly pruritus evaluation) with physician-assessed clinical signs into a unified assessment framework, addressing the current fragmentation between subjective and objective evaluation components. Third, this represents the AD assessment instrument developed through formal consensus methodology among Chinese dermatology experts, ensuring its relevance and applicability in the Chinese clinical context. These advances collectively address critical limitations in current AD assessment paradigms, particularly regarding clinical utility and implementation in real-world practice.

Despite its strengths, this study has several limitations. First, unlike existing validated assessment tools such as SCORAD and EASI, which were developed in Western populations, our newly developed scale incorporates expert insights from 16 dermatologists across China. Since clinical presentation and management of AD can vary across regions, the perspectives captured may not reflect global patterns. Therefore, the generalizability of findings is limited, and further studies incorporating international expert input are warranted to evaluate the scale’s global relevance.

Second, the expert panel expressed reservations regarding the exclusion of certain indicators. For instance, some panelists believed that frequent relapses (“D3 relapse history”) reflect inadequate treatment response and should be retained as a marker of disease severity. Similarly, regarding “D2 past systemic therapy response”, which includes responses to immunosuppressant (including herbal

medicine with the effect of immunosuppressant), glucocorticoid, biologics, molecularly targeted therapy, some experts emphasized its value in severe AD as it often cannot be controlled by topical treatments alone. Furthermore, systemic treatment misuse is less common than topical treatments; response to systemic treatments may have a stronger correlation with disease severity. Another debated item was “B2 location of skin lesion (referring to sensitive or visible parts of the body, including perineum, nipples, head and face, neck, hands. Symmetrical parts count as one, e.g., left and right hand only count as one)”. Although some of these areas are small in size, their impact on quality of life can be disproportionately large. As a result, certain experts argued for their retention.

In this study, these indicators were excluded solely based on the expert panel consensus during the Delphi process. Although expert consensus provides a valuable framework, these indicators could be important in the context of a broader clinical perspective. Therefore, it is recommended that these indicators be reconsidered in future research phases, potentially involving a large sample of clinical data and more comprehensive statistical analysis, such as calculating the weight of each indicator to determine its importance and relevance to the new scale.

Third, a well-constructed measurement scale, such as SCORAD and EASI, includes a comprehensive weighted scoring system based on the relative importance (weights) of its constituent items. This study did not generate indicator weighting for the Simplified AD Assessment Scale, which limits its precision. In future phases, we plan to explore several widely used methods to assign weights to each indicator:

(1) Analytic hierarchy process-entropy^[27-29] and superiority chart method,^[14] which rely on expert scoring of the importance of each indicator; (2) entropy method, which calculates weights using the data's information entropy, valuing variability; (3) Criteria Importance Through Intercriteria Correlation (CRITIC) and information-based weighting, which assign weights based on data variability and the relationships between indicators; and (4) factor and principal component analysis, which calculate weights based on the variance each factor explains in the data.

Although the analytic hierarchy process-entropy method was primarily based on expert judgment, the latter three are data driven. Each method has strengths and limitations in terms of balancing objectivity and contextual relevance. We aim to test multiple approaches in the second phase of the study to enhance the scale's robustness.

Fourth, the evaluation range for “skin lesion involvement of body surface area” (0–10%) may result in a “ceiling effect”,^[30] limiting its correlation with broader measures such as SCORAD or EASI. In the second phase of the project, we will optimize the evaluation range by including a wider sample of physicians and patients across diverse severity levels, enhancing the scale's accuracy and applicability, avoiding “ceiling effect”.

Finally, formal testing of the scale's reliability and validity has not yet been conducted. Existing scales such as SCORAD and EASI have all undergone such testing. This step is critical for confirming the tool's clinical utility and will be undertaken in the next phase of development.

Strengths and limitations of Simplified Criteria for Initiating AD Systemic Treatment

Existing guidelines for initiating systemic therapy in AD often lack clarity and clinical practicality, which may delay appropriate treatment. For example, the 2020 Chinese guideline recommends systemic therapy based on SCORAD thresholds, which can be time-consuming and complex.^[3] Similarly, in expert recommendations on treat-to-target in the systemic treatment of moderate to severe AD,^[4] patients should meet at least one of the following three criteria to be considered systemic treatment: objective scores (e.g., EASI, PGA, SCORAD), subjective assessments (e.g., DLQI, VAS, NRS), and evidence of treatment failure. Portuguese recommendations also require multiple indicators before systemic therapy is considered^[9]: EASI >16, or EASI <16 who meet one of the following, (1) localized on face, hands, or genitals; (2) pruritus NRS >7; (3) sleep disturbances with NRS >7; and (4) DLQI >10. As can be seen, although structured, these approaches are complex, potentially delaying treatment initiation or leading to inconsistent criteria across different clinicians.

In contrast, our proposed criteria were developed through expert consensus and two rounds of eDelphi surveys. These criteria consider these dimensions: body surface area skin lesion involvement, special area involvement, past topical therapy response, and impacts on sleep, work, study and socializing. This

approach streamlines clinical decision-making by eliminating complex calculations, improving usability, enabling timely treatment initiation and reducing risks of both over- and undertreatment.

However, limitations remain. First, definitions of topical therapy failure are inconsistent across settings, especially in primary care. Second, lesion characteristics (e.g., erythema, papulation) were excluded, potentially overlooking important severity markers. Third, reliance on subjective symptoms (e.g., itch, sleep) may introduce assessment variability. Last, the criteria rely solely on China expert opinions. It has not yet been validated in clinical practice and therefore the sensitivity and specificity remains unclear.

Future directions: optimization, reliability testing and validity testing of the Simplified AD Assessment Scale

In the second phase of our project, we plan to involve a substantial number of physicians and patients to evaluate the Simplified AD Assessment Scale at baseline (week 0), and at weeks 2, 4, 8, and 12. Concurrently, established instruments such as SCORAD, EASI, DLQI, POEM, and PP-NRS will be used as benchmarks. Through statistical analyses of assessments from both physicians and patients, we aim to refine the scale's structure and descriptions, assigning appropriate weights to each item. Furthermore, we will validate the scale's reliability, including internal consistency and test-retest reliability, and its validity, encompassing content and criterion validity.^[31] Rigorous reliability and validity testing are essential for the Simplified AD Assessment Scale before clinical application. Dermatological scales like EASI,^[32] the 5-D pruritus scale,^[33] PP-NRS,^[34] Validated Investigator Global Assessment for Atopic Dermatitis (vIGA-AD),^[35] IGA × BSA,^[36] Itch Severity Scale (ISS)^[37] have all undergone such testing post-development, with reliability and validity assessed separately.^[38] Without these evaluations, the scale's clinical significance cannot be assured, posing challenges in managing AD. In addition, the second phase of the study will use the Simplified Assessment Scale to design a treat-to-target framework for AD, aiding clinical decision-making and disease management.

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Conflicts of interest

None.

References

1. Zhao J, Zhang Z, Chen H, Dou X, Zhao Z, Liu L, et al. Associations of demographics, aggravating factors, comorbidities, and treatments with atopic dermatitis severity in China: A national cross-sectional study. *Chin Med J* 2024;138:553–561. doi: 10.1097/cm9.0000000000003042.

- [Cited Here](#) |
- [Google Scholar](#)

2. Hadi HA, Tarmizi AI, Khalid KA, Gajdács M, Aslam A, Jamshed S. The epidemiology and global burden of atopic dermatitis: A narrative review. *Life (Basel)* 2021;11:936. doi: 10.3390/life11090936.

- [Cited Here](#) |
- [Google Scholar](#)

3. Xu Y, Qiang SZ, Wei L, Sheng LY, Yan Z, Hua C, et al. Guidelines for the diagnosis and treatment of atopic dermatitis in China (2020): The Chinese Society of Dermatology Immunology Group and Collaborative Research Center of Atopic Dermatitis. *Int J Dermatol Venereol* 2021;53:81–88. doi: 10.35541/cjd.20191000.

- [Cited Here](#) |
- [Google Scholar](#)

4. Zhao Z, Gao X. Expert recommendations on treat-to-target in the systemic treatment of moderate to severe atopic dermatitis (in Chinese). *Chin J Dermatovenereol* 2022;36:855–864. doi: 10.13735/j.cjdv.1001-7089.202205115.

- [Cited Here](#) |
- [Google Scholar](#)

5. De Bruin-Weller M, Biedermann T, Bissonnette R, Deleuran M, Foley P, Girolomoni G, et al. Treat-to-target in atopic dermatitis: An International Consensus on a Set of Core Decision Points for Systemic Therapies. *Acta Derm Venereol* 2021;101:adv00402. doi: 10.2340/00015555-3751.

- [Cited Here](#) |
- [Google Scholar](#)

6. Severity scoring of atopic dermatitis: The SCORAD index. Consensus Report of the European Task Force on Atopic Dermatitis. *Dermatology* 1993;186:23–31. doi: 10.1159/000247298.

- [Cited Here](#) |
- [Google Scholar](#)

7. Tofte S, Graeber M, Cherill R, Omoto M, Thurston M, Hanifin JM. Eczema area and severity index (EASI): A new tool to evaluate atopic dermatitis. *J Eur Acad Dermatol Venereol* 1998;11:S197. doi: 10.1016/S0926-9959(98)95291-6.

- [Cited Here](#) |
- [Google Scholar](#)

8. Kamp E, Ascott A, George SMC. Eczema severity scoring in skin of color: A review of current best practice and need for future improvement. *J Invest Dermatol* 2025;145:735–748. doi: 10.1016/j.jid.2025.02.001.

- [Cited Here](#) |
- [Google Scholar](#)

9. Torres T, Gonçalo M, Paiva Lopes MJ, Claro C, Ramos L, Selores M, et al. Portuguese recommendations for the treatment of atopic dermatitis with biologic therapy and JAK inhibitors in adult patients. *Drugs Context* 2021;10:2021-9-5. doi: 10.7573/dic.2021-9-5.

- [Cited Here](#) |
- [Google Scholar](#)

10. Strober B, Ryan C, van de Kerkhof P, van der Walt J, Kimball AB, Barker J, et al. Recategorization of psoriasis severity: Delphi consensus from the International Psoriasis Council. *J Am Acad Dermatol* 2020;82:117–122. doi: 10.1016/j.jaad.2019.08.026.

- [Cited Here](#) |
- [Google Scholar](#)

11. Harold L, Murray T. *The Delphi Method: Techniques and Applications*. Addison-Wesley, 1975.

- [Cited Here](#)

12. Yang J, Wang MH, Li JS, Lin XH, Liu YY, Xie Y, et al. Risk prediction questionnaire and item screening for acute exacerbation of asthma based on the modified Delphi technique (in Chinese). *Chin Gen Practice* 2022;25:4425–4432. doi: 10.12114/j.issn.1007-9572.2022.0489.

- [Cited Here](#) |
- [Google Scholar](#)

13. Wu YX, Liu XH, Wu YJ, Zhang YJ, Zhang XH, Hou BB, et al. Development and validation of a symptom assessment scale for patients with non-small cell lung cancer treated with immune checkpoint inhibitors (in Chinese). *Chin Nursing Res* 2023;37:826–832. doi:10.12102/j.issn.1009-6493.2023.05.011.

- [Cited Here](#) |
- [Google Scholar](#)

14. Yao J, Yuan JQ, Ma LY, Liang ZP. Construction of a risk assessment indicator system for ICU-acquired weakness based on the Delphi method (in Chinese). *Lab Med Clin* 2023;20:2041–2044. doi: 10.3969/j.issn.1672-9455.2023.14.012.

- [Cited Here](#) |
- [Google Scholar](#)

15. Zhu XL, Ma TH, Qi XF, Wang YL, Feng KX, Liu X. Construction of perioperative anesthesia care quality sensitive indicators based on evidence-based and Delphi methods (in Chinese). *Int J Anesthesiol Res* 2022;43:956–962. doi: 10.3760/cma.j.cn321761-20220412-00634.

- [Cited Here](#) |
- [Google Scholar](#)

16. Wang SQ, Liang WX. Construction of an efficacy evaluation system for acupuncture treatment of ischemic stroke in the recovery period based on expert survey method (in Chinese). *J Yunnan Univ TCM* 2014;37:42–46. doi: CNKI:SUN:YNZY.0.2014-03-015.

- [Cited Here](#) |
- [Google Scholar](#)

17. Shi C, Zhang Y, Li C, Li P, Zhu H. Using the Delphi method to identify risk factors contributing to adverse events in residential aged care facilities. *Risk Manag Healthc Policy* 2020;13:523–537. doi: 10.2147/rmhp.S243929.

- [Cited Here](#) |
- [Google Scholar](#)

18. Niu G, Chen H, Liu L, Zhou G, Zhou Q, Li C, et al. Surgical treatment for upper cervical deformity with atlantoaxial joint dislocation using individualized 3D printing occipitocervical fusion instrument: A case report and literature review. *Medicine (Baltimore)* 2021;100: e25202. doi: 10.1097/md.00000000000025202.

- [Cited Here](#) |
- [Google Scholar](#)

19. Yan L, Pu H, Shi X, Yu Q, Wang Z, Hu Y, et al. Construction of key quality indicators for aged care facilities in China: A two-tier Delphi study: Key aged care quality indicators in ACF by Delphi method. *J Clin Nurs* 2024;33:752–767. doi: 10.1111/jocn.16917.

- [Cited Here](#) |
- [Google Scholar](#)

20. Chu DK, Schneider L, Asiniwasis RN, Boguniewicz M, De Benedetto A, et alAAAAI/ACAAI JTF Atopic Dermatitis Guideline Panel. Atopic dermatitis (eczema) guidelines: 2023 American Academy of Allergy, Asthma and Immunology/American College of Allergy, Asthma and Immunology Joint Task Force on Practice Parameters GRADE- and Institute of Medicine-based recommendations. *Ann Allergy Asthma Immunol* 2024;132:274–312. doi: 10.1016/j.anai.2023.11.009.

- [Cited Here](#) |
- [Google Scholar](#)

21. Committee on Allergic Diseases; China Dermatologist Association; Atopic Dermatitis Research Center; Chinese Society of Dermatology; Dermatology Branch of China International Exchange and Promotion Association for Medical and Healthcare. Expert consensus for management of itch in atopic dermatitis(in Chinese). *Chin J Dermatol* 2021;54:391–396. doi:10.35541/cjd.20210056.

- [Cited Here](#) |
- [Google Scholar](#)

22. Xu Y. How to evaluate the severity of atopic dermatitis (in Chinese)? *Chin J Dermatol* 2021;54:539–541. doi: 10.35541/cjd.20200825.

- [Cited Here](#) |
- [Google Scholar](#)

23. Eichenfield LF, Tom WL, Berger TG, Krol A, Paller AS, Schwarzenberger K, et al. Guidelines of care for the management of atopic dermatitis: Section 2. Management and treatment of atopic dermatitis with topical therapies. *J Am Acad Dermatol*. 2014;71:116–132. doi: 10.1016/j.jaad.2014.03.023.

- [Cited Here](#) |
- [Google Scholar](#)

24. Wollenberg A, Kinberger M, Arents B, Aszodi N, Avila Valle G, Barbarot S. European guideline (EuroGuiDerm) on atopic eczema: Part I - Systemic therapy. *J Eur Acad Dermatol Venereol* 2022;36:1409–1431. doi: 10.1111/jdv.18345.

- [Cited Here](#) |
- [Google Scholar](#)

25. Sidbury R, Alikhan A, Bercovitch L, Cohen DE, Darr JM, Drucker AM, et al. Guidelines for management of atopic dermatitis. *J Am Acad Dermatol* 2010;36:563–577. doi: 10.1016/j.jaad.2022.12.029.

- [Cited Here](#) |
- [Google Scholar](#)

26. Chen QQ, Yang XJ, Gu H, Xu JH, Hao F, Yao X, et al. Construction of “Expert consensus on diagnosis and treatment of chronic inducible urticaria in China (2023)” based on the Delphi method (in Chinese). *Chin J Dermatol* 2023;56:534–539. doi: 10.35541/cjd.20230216.

- [Cited Here](#) |
- [Google Scholar](#)

27. Yagupsky P, Morata P, Colmenero JD. Laboratory diagnosis of human brucellosis. *Clin Microbiol Rev* 2019; 33:e00073–19. doi: 10.1128/CMR.00073-19.

- [Cited Here](#) |
- [Google Scholar](#)

28. Liu X, Zhang W, Qu Z, Guo T, Sun Y, Rabiei M, et al. Feasibility evaluation of hydraulic fracturing in hydrate-bearing sediments based on analytic hierarchy process-entropy method (AHP-EM). *J Nat Gas Sci Eng* 2020;81:103434. doi: 10.1016/j.jngse.2020.103434.

- [Cited Here](#) |
- [Google Scholar](#)

29. Wu SW, Chen T, Pan Q, Wei LY, Wang Q, Li C, et al. Establishment of a quantitative medical technology evaluation system and indicators within medical institutions. *Chin Med J* 2018;131:1327–1332. doi: 10.4103/0366-6999.232804.

- [Cited Here](#) |
- [Google Scholar](#)

30. Klapproth CP, Fischer F, Rose M. Outcomes, scale agreement, ceiling and floor effects, construct validity, and relative efficiency of the PROPr and EQ-5D-3L in low back pain patients. *Health Qual Life Outcomes* 2023;21:107. doi: 10.1186/s12955-023-02188-w.

- [Cited Here](#) |
- [Google Scholar](#)

31. Nan Yang, Hui Liu, Wei Zhao, Yang Pan, Xiangzheng Lyu, Xiuyuan Hao, et al. Development of the Scientific, Transparent and Applicable Rankings (STAR) tool for clinical practice guidelines. *Chin Med J* 2023;136:1430–1438. doi: 10.1097/CM9.0000000000002713.

- [Cited Here](#) |
- [Google Scholar](#)

32. Hanifin JM, Thurston M, Omoto M, Cherill R, Tofte SJ, Graeber M. The eczema area and severity index (EASI): Assessment of reliability in atopic dermatitis. EASI Evaluator Group. *Exp Dermatol* 2001;10:11–8. doi: 10.1034/j.1600-0625.2001.100102.x.

- [Cited Here](#) |
- [Google Scholar](#)

33. Elman S, Hynan LS, Gabriel V, Mayo MJ. The 5-D itch scale: A new measure of pruritus. *Br J Dermatol* 2009;162: 587–593. doi: 10.1111/j.1365-2133.2009.09586.x.

- [Cited Here](#) |
- [Google Scholar](#)

34. Yosipovitch G, Reaney M, Mastey V, Eckert L, Abbé A, Nelson L, et al. Peak Pruritus Numerical Rating Scale: psychometric validation and responder definition for assessing itch in moderate-to-severe atopic dermatitis. *Br J Dermatol* 2019;181:761–769. doi: 10.1111/bjd.17744.

- [Cited Here](#) |
- [Google Scholar](#)

35. Simpson E, Bissonnette R, Eichenfield LF, Guttman-Yassky E, King B, Silverberg JI, et al. The Validated Investigator Global Assessment for Atopic Dermatitis (vIGA-AD): The development and reliability testing of a novel clinical outcome measurement instrument for the severity of atopic dermatitis. *J Am Acad Dermatol* 2020;83:839–846. doi: 10.1016/j.jaad.2020.04.104.

- [Cited Here](#) |
- [Google Scholar](#)

36. Suh TP, Ramachandran D, Patel V, Jackson KL, Rangel SM, Fishbein AB, et al. Product of Investigator Global Assessment and Body Surface Area (IGAxBSA): A practice-friendly alternative to the Eczema Area and Severity Index to assess atopic dermatitis severity in children. *J Am Acad Dermatol* 2020;82:1187–1194. doi: 10.1016/j.jaad.2019.12.019.

- [Cited Here](#) |
- [Google Scholar](#)

37. Majeski CJ, Johnson JA, Davison SN, Lauzon CJ. Itch Severity Scale: A self-report instrument for the measurement of pruritus severity. *Br J Dermatol* 2010;156:667–673. doi: 10.1111/j.1365-2133.2006.07736.x.

- [Cited Here](#) |
- [Google Scholar](#)

38. Brundle C, Heaven A, Brown L, Teale E, Young J, West R, et al. Convergent validity of the electronic frailty index. *Age Ageing* 2019;48:152–156. doi: 10.1093/ageing/afy162.

- [Cited Here](#) |
- [Google Scholar](#)

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